Evolution of general health indicators

Group 33: Duarte Carvalhal (20221820) | Duarte Neves (20221880) | Filipe Flaminio (20221813)

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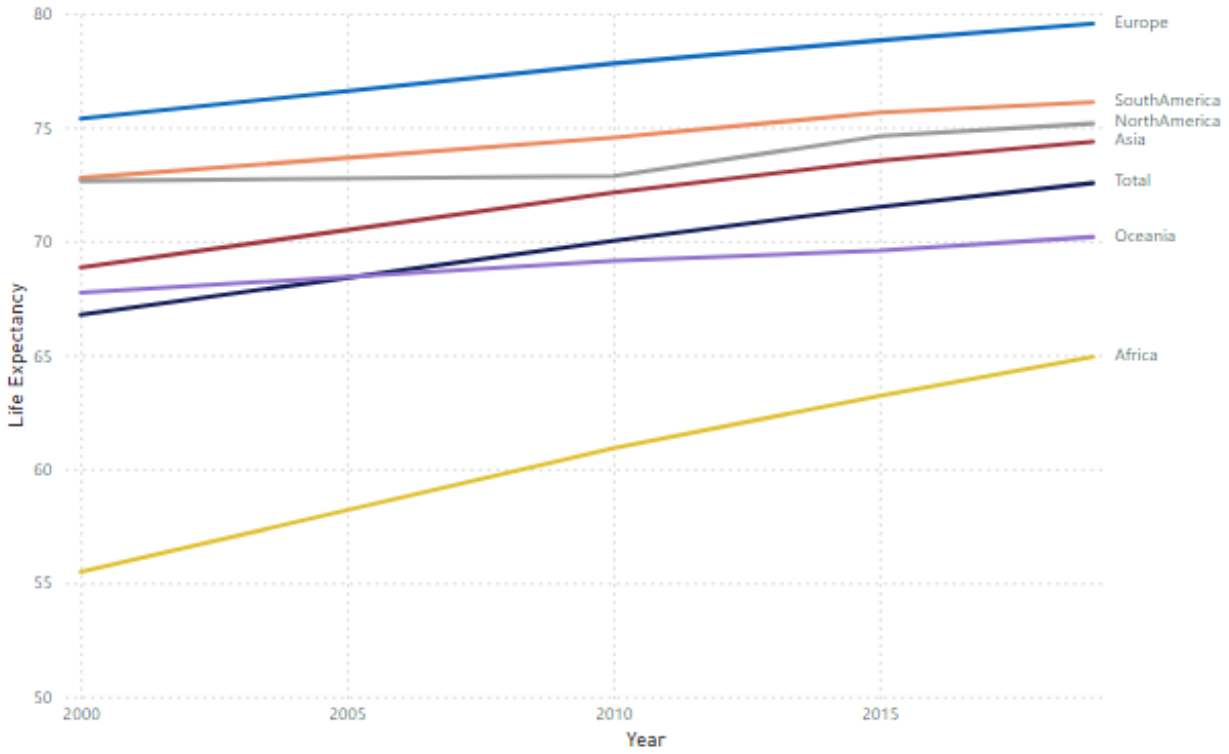
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Life expectancy

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Average Life Expenctancy in 2019: 70 years



Positive growth trend in average life expectancy

Overall life expectancy of the whole world is increasing severely, showing that the medical discoveries, technologies and advances are being incredibly effective.

Developed countries with higher values

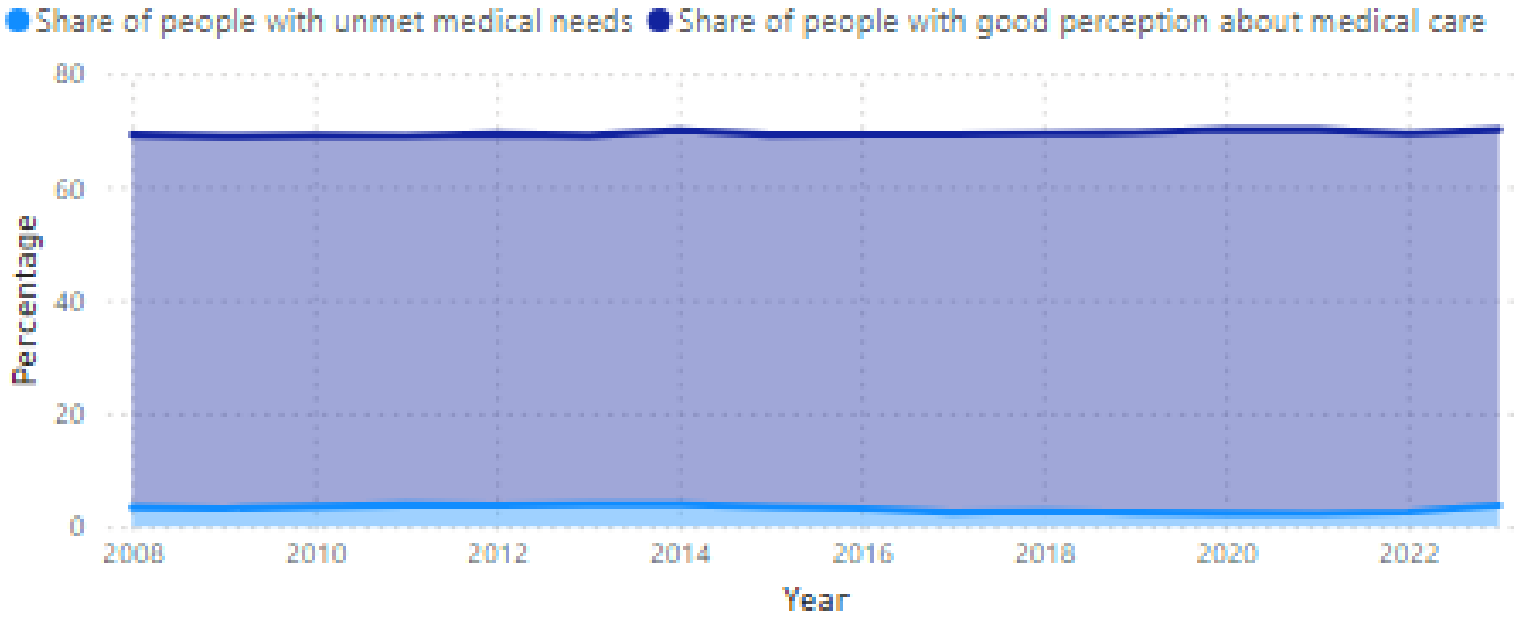
Countries with better social and economic conditions have higher average life expectancy rates due to their strong development in the health area

Data taken from: World Health Organization

Perception about health in the EU

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Most Europeans have a good perception of medical care



In 2022, the percentage of people that have a good perception about the medical care in the EU (67%), we can also see that, overtime, this value is stabilized, having minimal oscillations. On the other side we have the percentage of people with unmet medical needs that oscillates between two and four percent, it can be concluded that, for the European Union, these figures are somewhat satisfying.

Data taken from: eurostat

Mortality Rate Under 5

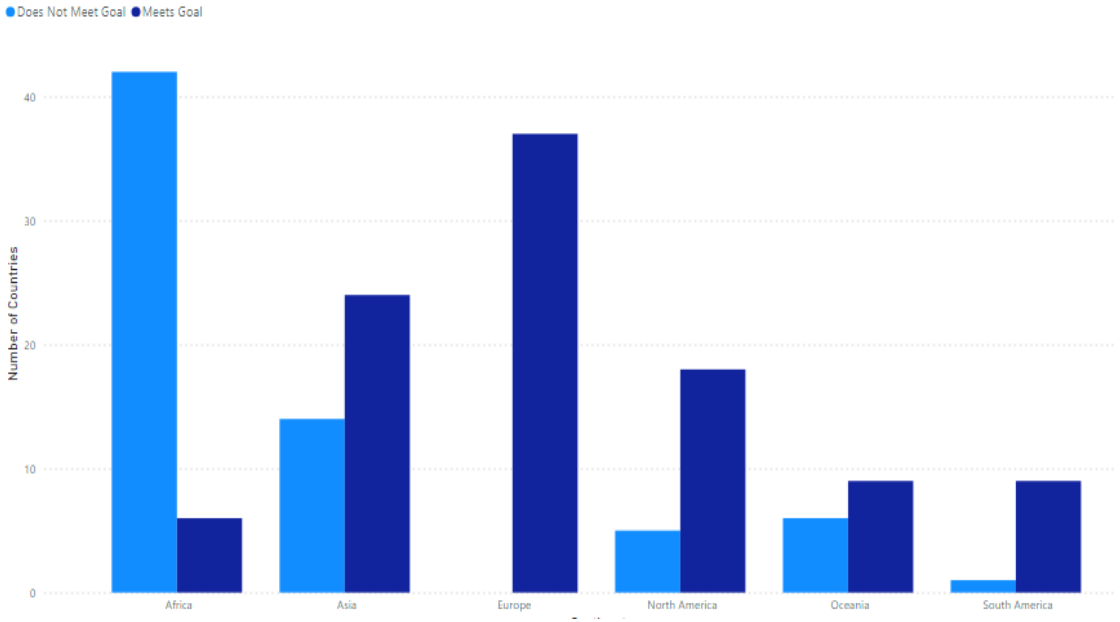
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SDG Target 3.1

1. Neonatal Mortality Rate: The UN's objective for neonatal mortality is to reduce the neonatal mortality rate to at least as low as 12 per 1,000 live births or lower by 2030.

2. Under-5 Mortality Rate: The UN aims to end preventable deaths of newborns and children under 5 years of age, with all countries aiming to reduce under-5 mortality to at least as low as 25 per 1,000 live births or lower by 2030.

Countries meeting the SDG Target 3.1 in 2019



Neonatal and under-5 deaths fall steadily

Neonatal and under-5 deaths have declined steadily, with all European countries meeting UN objective 3.1 for sustainable development. On the other hand, the African continent is a victim of its poor access to basic health facilities and resources.

Data taken from: kaggle

Maternal Mortality Rate

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Only Europe and Oceania are already below the SDG 3.1 target

Maternal Mortality around the world



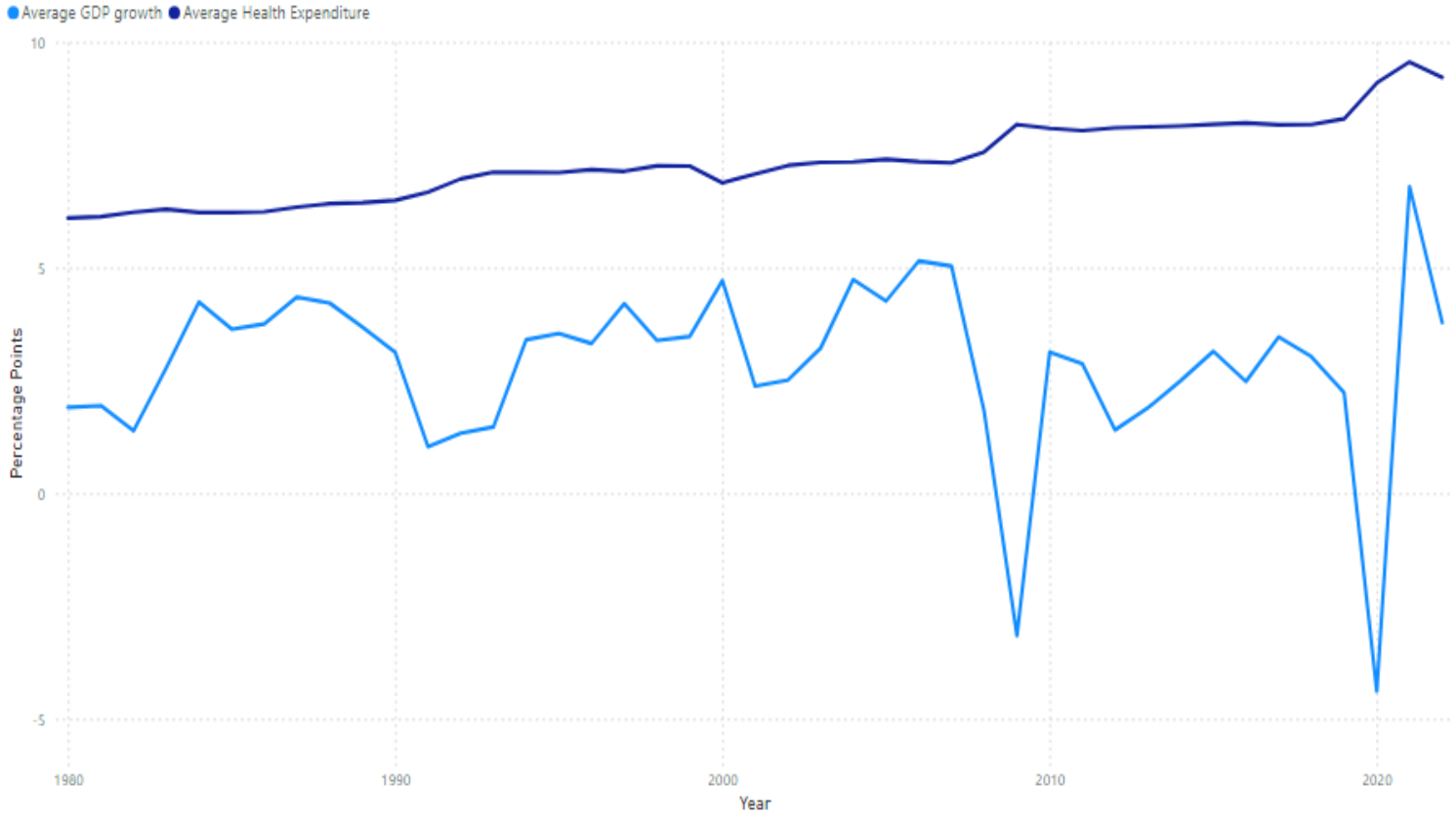
The UN's objective for maternal mortality (SDG 3.1) is to reduce the global maternal mortality ratio (MMR) to less than 70 per 100,000 live births by 2030.
Overall, it is still far from reaching the SDG 3.1 target, even with, in 2015, 65% of countries already meeting the target, this is due to the lack of medical conditions and monitoring capacity in less deprived countries, mainly on the African continent. .

Data taken from: kaggle

Health Expenditure by OECD Countries

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OECD countries increase health spending to combat Covid-19



Health expenditure immune to fluctuations in GDP

Besides these fluctuations and dips we can see a big effort by most countries to increase the percentage of the GDP spent on health even, and especially, in periods of crisis. In 2009 the average country met the goal of having their health expenditure >8%, and this figure keeps increasing until today.

Commitment to fighting the pandemic

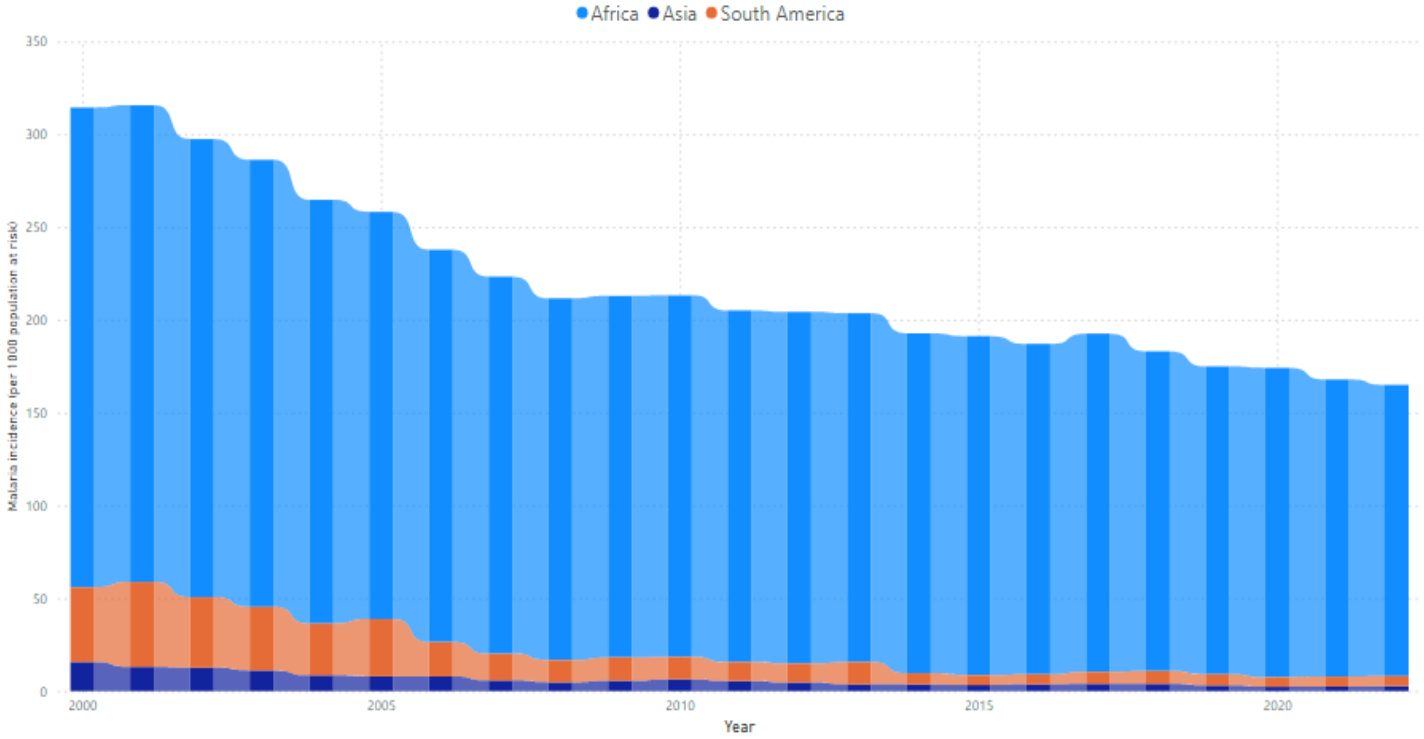
All OECD countries increased their share of healthcare spending as a way to combat the Covid-19 pandemic that began at the beginning of 2020.

Data taken from: OECD

Estimated Malaria Incidence

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Estimated malaria incidence (per 1000 population at risk)



Malaria in no longer a threat in continents like Europe and North America

But, in some South American and Asian countries and specially in Africa efforts are being made to fight this troublesome decease, looking at the graph we can see that the main incidence is by far in Africa, we can also see that it has decayed 50% in the past 20 years.

Even so, the estimated value of the incidence of malaria per 1000 people at risk was 156.9 in Africa, 2.75 in Asia and 5.33 in South America in 2022.

Data taken from: World Health Organization

Hepatitis B prevalence

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Hepatitis B prevalence among children under 5 years (%) in 2020

World	Africa	Asia	Europe	North America	South America	Oceania
0,89	2,38	0,30	0,26	0,19	0,66	0,89

Hepatitis B under control around the world

More than 90% of the world's population has an almost zero or low prevalence, which means that hepatitis B is a problem almost solved throughout the world, this is the result of the success of public policies, many of which are implemented at an international level, the only relevant prevalence that still exists is in some African countries.

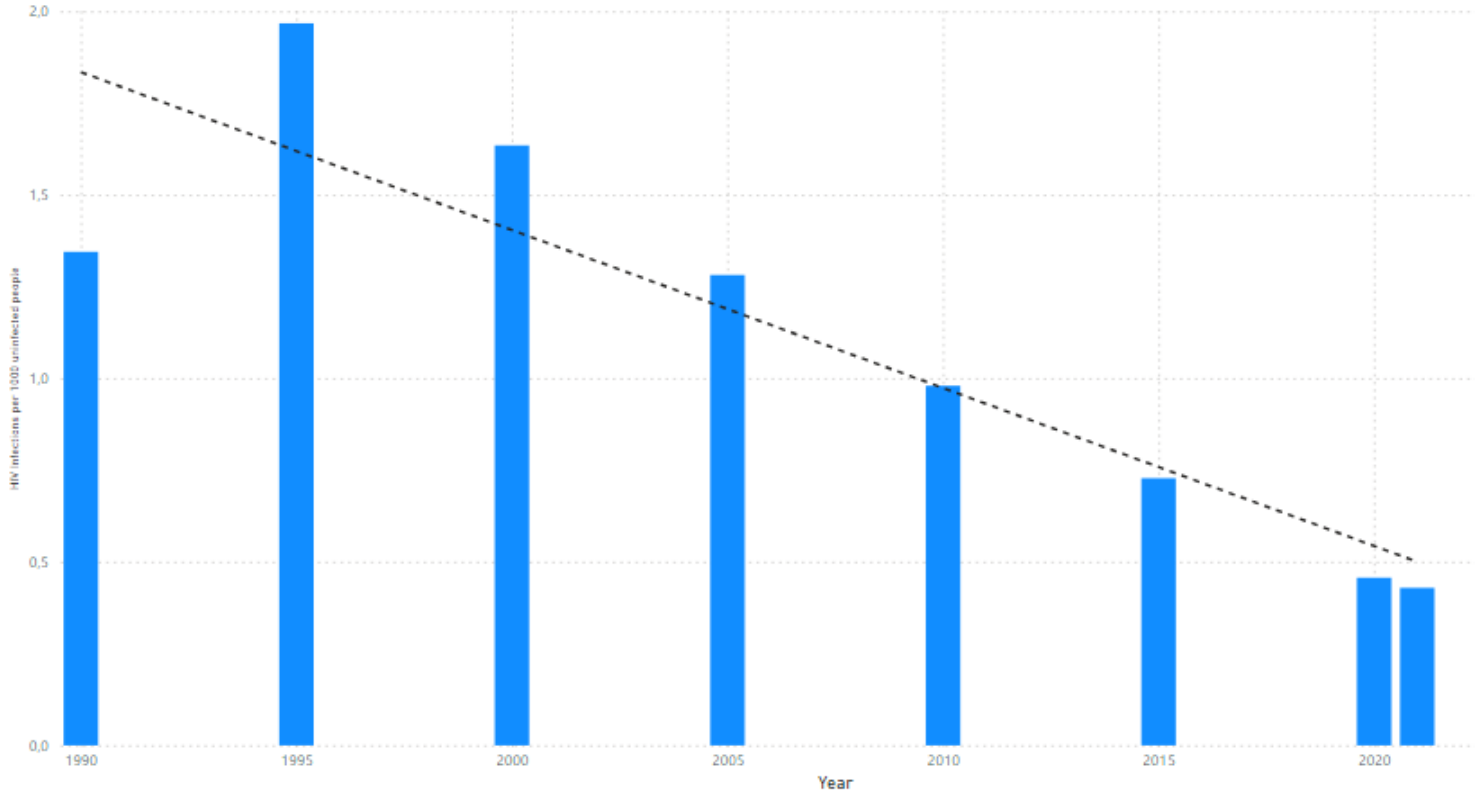


Data taken from: World Health Organization

New HIV infections

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New HIV infections per 1000 uninfected people



The UN's target for new HIV infections under SDG 3 reflects a comprehensive approach to combatting the HIV/AIDS epidemic by focusing on prevention, treatment, and care.

Decreasing trend in the number of new HIV infections

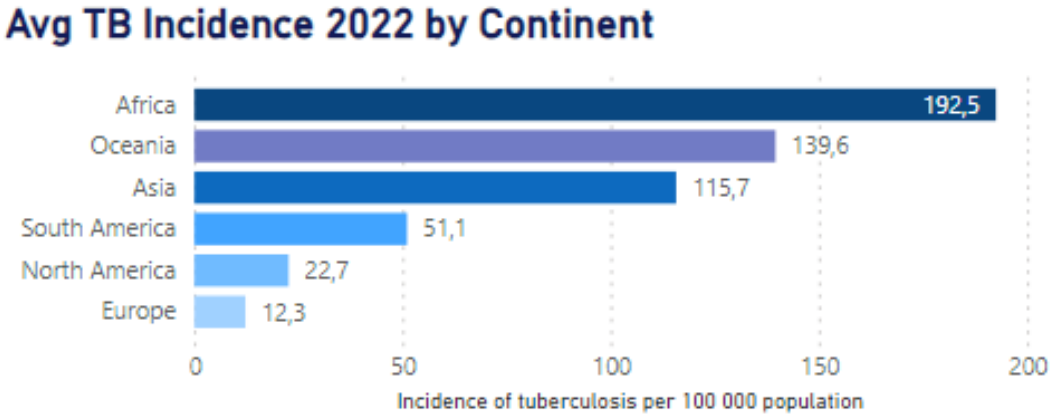
HIV is a disease that has been steadily decreasing in recent years due to efforts in medical studies and awareness actions to increase awareness about the disease and over the years the situation is mostly under control and decreasing rapidly, following this trend, In the future, HIV will no longer be a problem in our society.

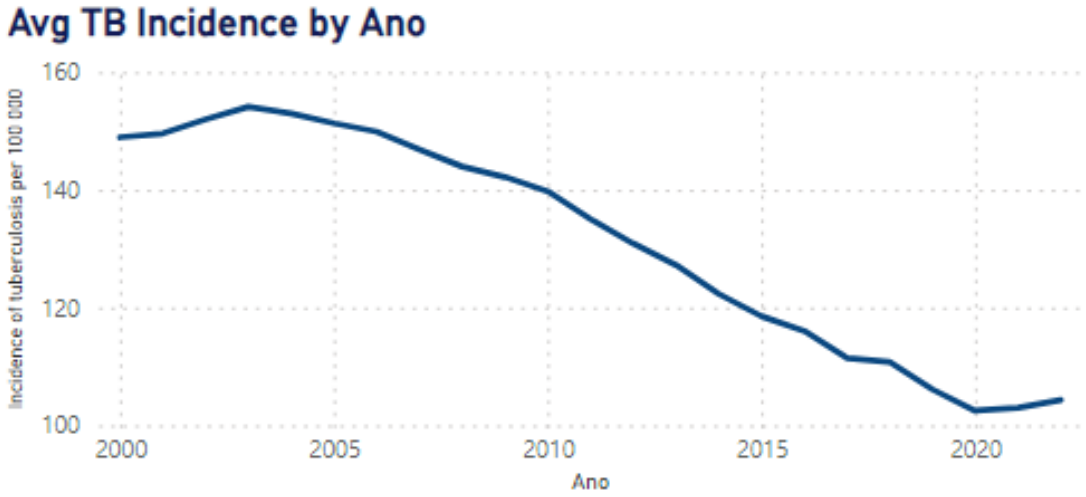
In 2021, close to 50% of countries will have a number of new HIV infections of 0.03 per 1000 uninfected people

Data taken from: World Health Organization

Incidence of tuberculosis

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The United Nations (UN) has established targets related to tuberculosis (TB) as part of its Sustainable Development Goals (SDGs). The specific target for TB falls under SDG 3.3 and reflects a comprehensive approach to TB control and prevention, emphasizing the importance of integrated, evidence-based interventions, multisectoral collaboration, and sustainable financing to achieve the goal.

Tuberculosis increases again with Covid-19 pandemic

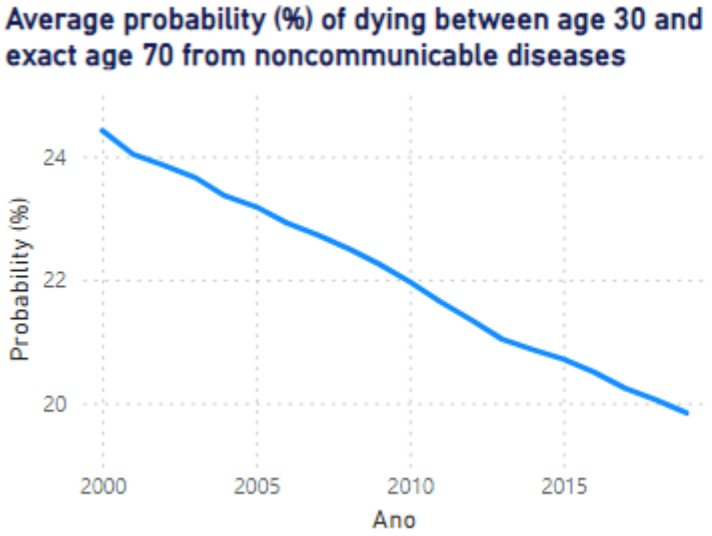
Since 2005 there has been a global effort to reduce the incidence of tuberculosis, and this effort has been accompanied by good results until 2020, where the Covid-19 pandemic brought additional challenges and compromised the downward trend in the number of tuberculosis cases. Africa, due to its weak infrastructure and health support, presents the highest value in 2022.

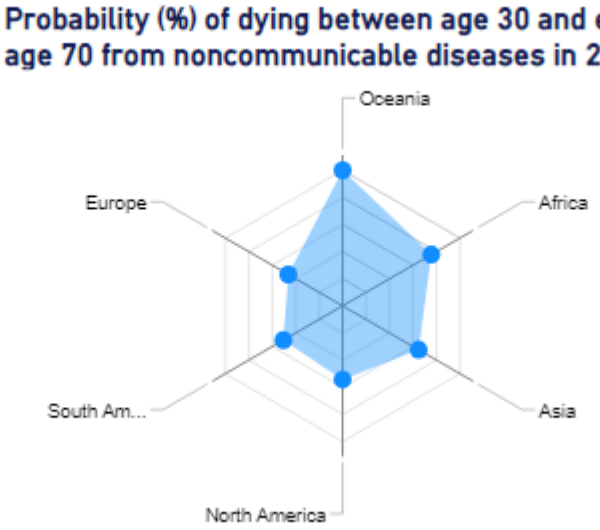
Data taken from: World Health Organization

Probabiblity of dying from Noncommunicable diseases

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The United Nations (UN) has set a target under SDG 3 to reduce premature mortality from NCDs by one third by 2030. This goal underscores the importance of prevention, treatment, and promoting mental health and well-being.





Average Probability in 2019

19,84

Variation from 2000 to 2019

-9,23%

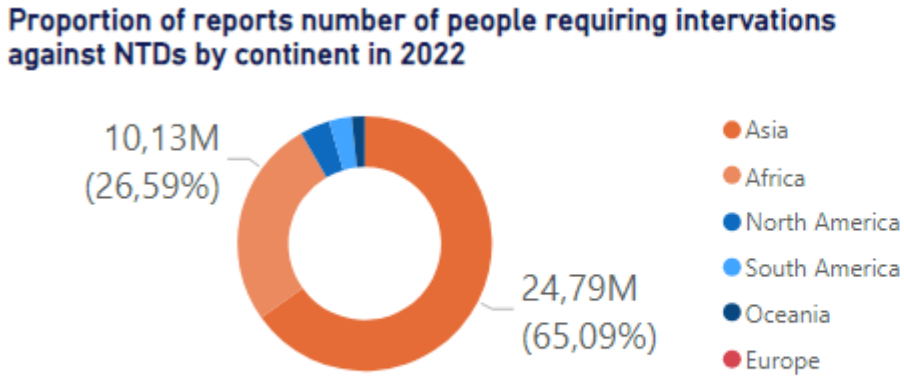
Reduction of one tenth in the probability of dying from non-communicable diseases

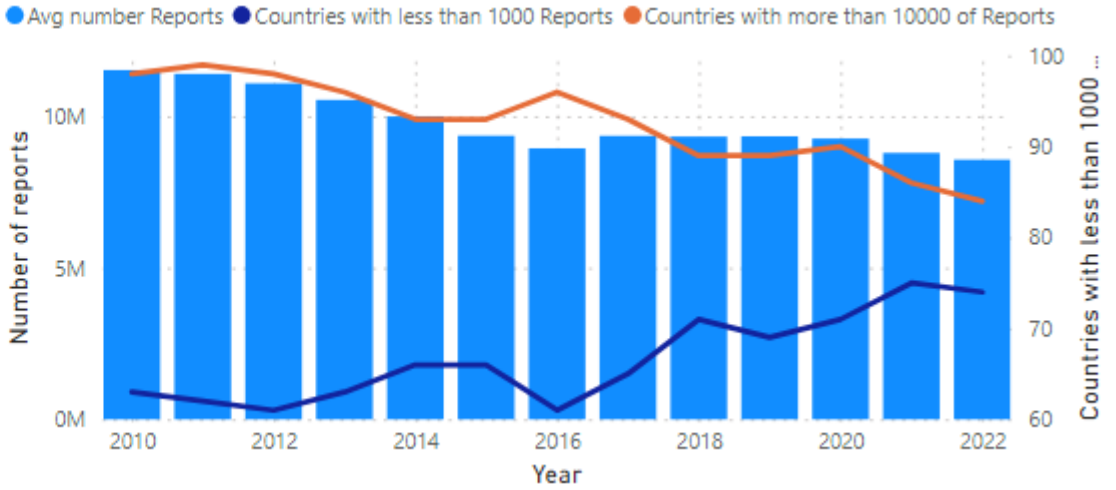
The average probability of dying between age 30 and exactly age 70 from noncommunicable diseases in 2019 was 19.84%, this figure since 2019 has reduced by around 9.23%, still short of the UN's target of reducing a third of this number by 2030. Even so, the trend has been decreasing, with Oceania and Africa having the highest values in terms of probability of dying between age 30 and exact age 70 from noncommunicable diseases in 2019.

Data taken from: World Health Organization

Reported number of people requiring interventions against NTDs

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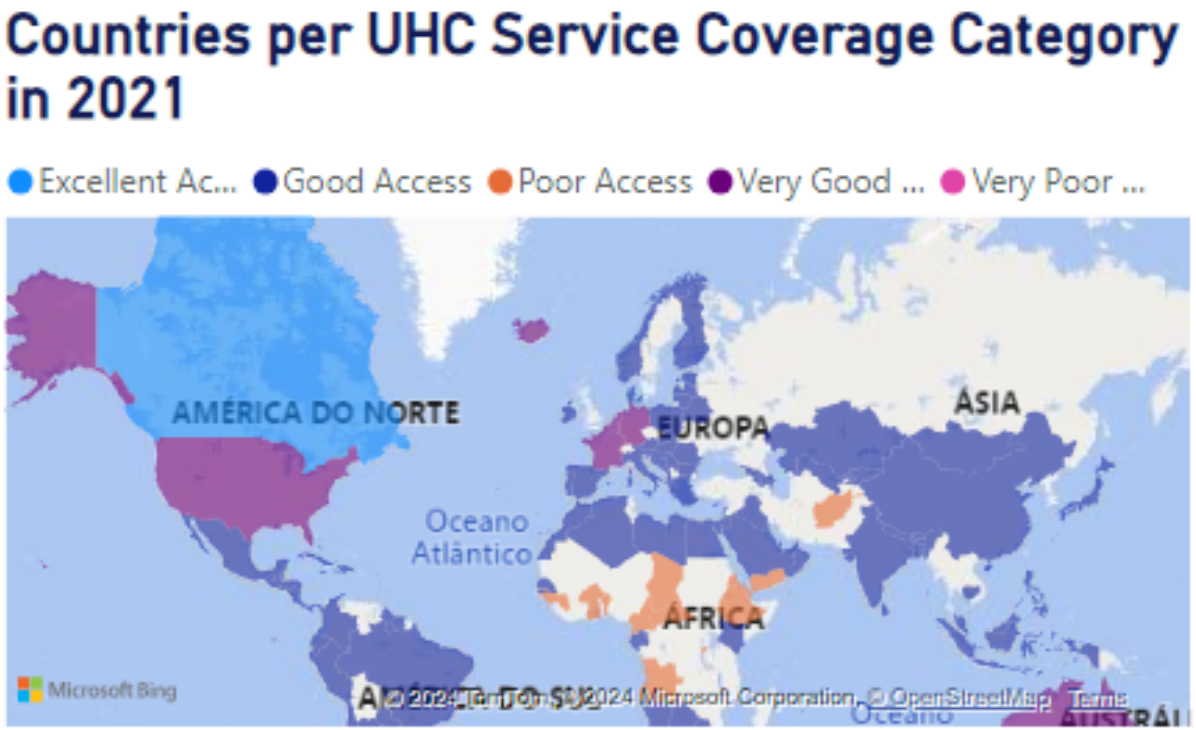
Variation of the number of people requiring interventions against NTDs was -21,57%

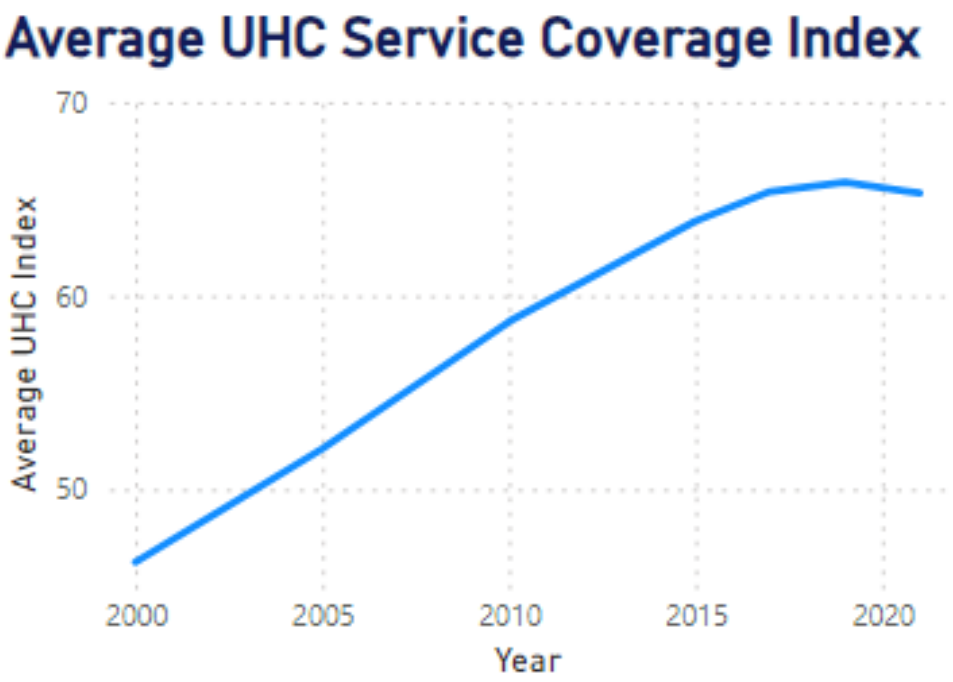
Globally, the value of the Reported number of people requiring interventions against NTDs has decreased by 21.57% since 2010 (until 2022), with Europe surpassing the UN target of reducing this value by 90% by 2030. Since 2010 we have been able to observe a decrease in the number of countries with many reports. This has been the trend in recent years despite the slight increase in the number of reports from 2016 to 2017.

Data taken from: World Health Organization

UHC Service Coverage Index

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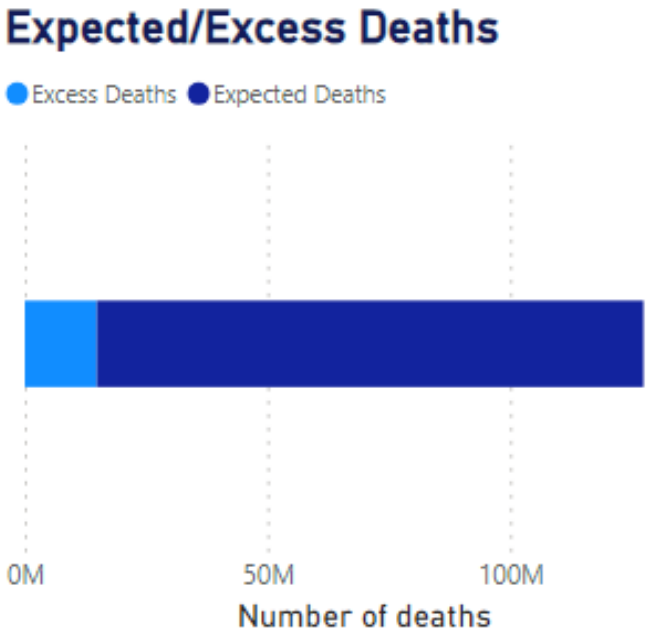
"Northern Countries" with better access to healthcare

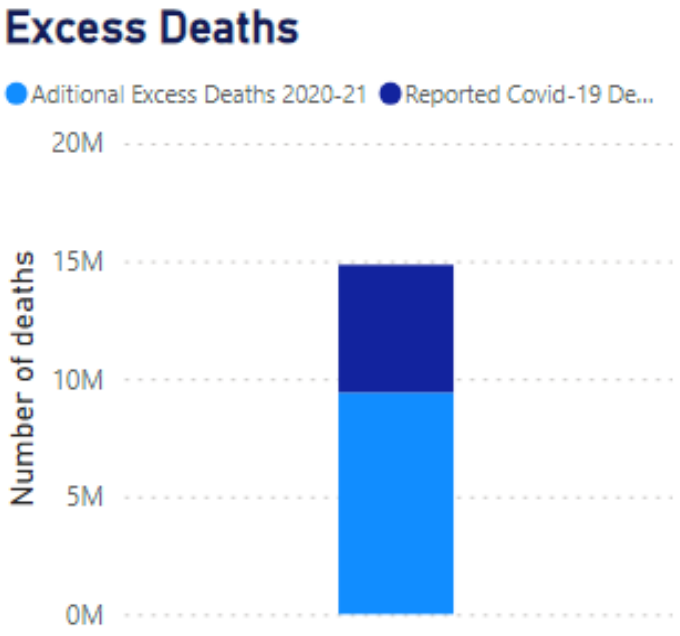
The countries called "Northern Countries" are those that have the best coverage rates in terms of health services in 2021, while the African continent, especially in the center, has the worst rates. This indicator (UHC Service Coverage Index) has been growing globally around the world until 2020 where with the start of the pandemic this indicator showed a slight decrease.

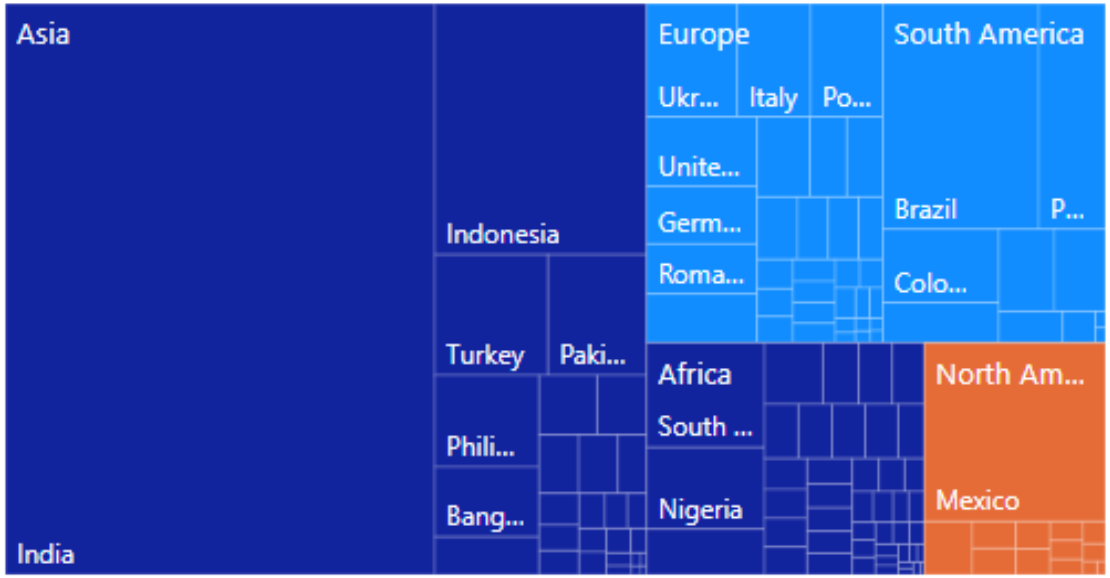
Data taken from: World Health Organization

The impact of COVID-19 - Excess of mortality in 2020-21

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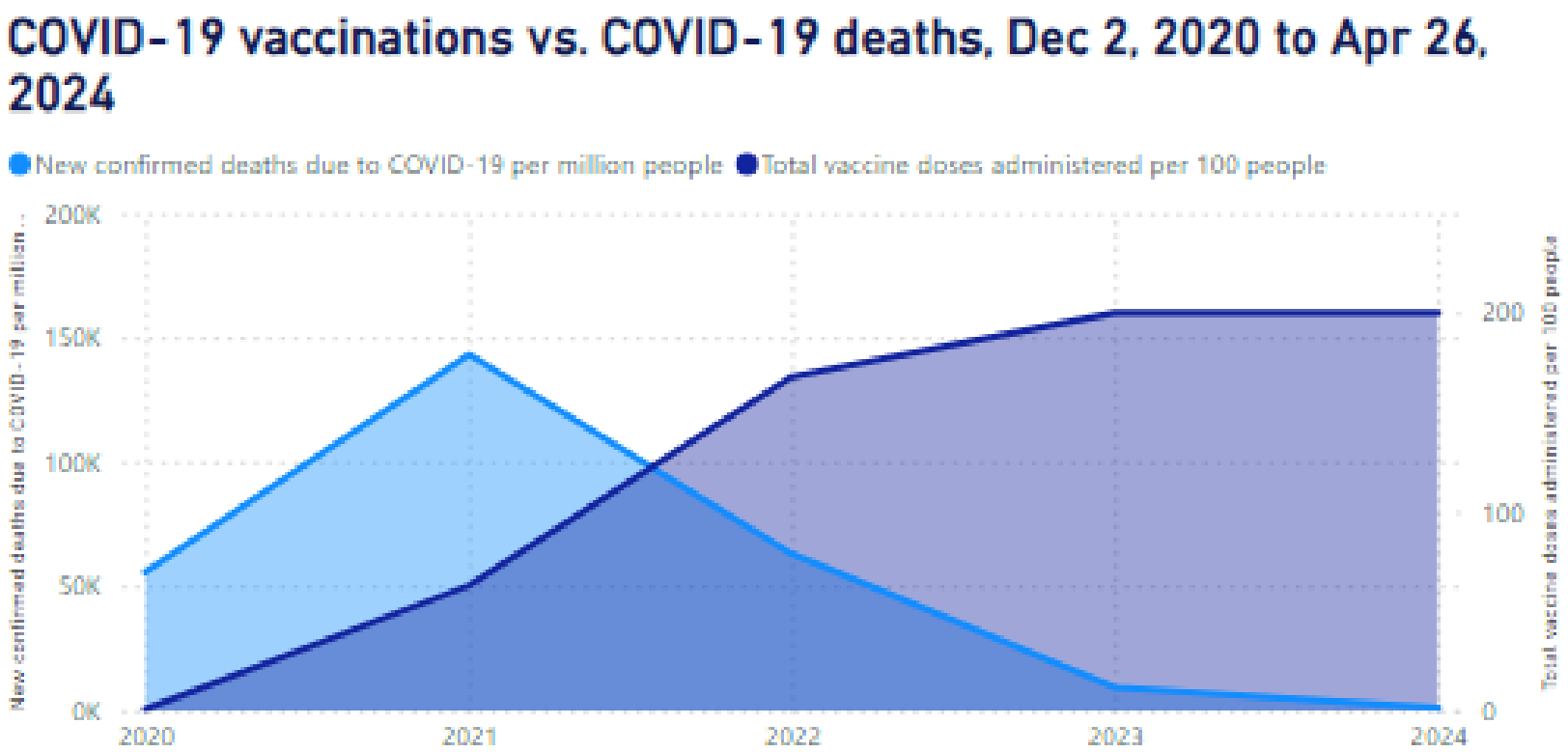
Covid-19 leads to a significant increase in the number of deaths.

The World Health Organization (WHO) has calculated excess mortality during the first two years of the pandemic. Approximately 15 million more people died in 2020 and 2021 (combined) than had been expected before the pandemic. Six countries (India, Russia, Indonesia, United States, Brazil, and Mexico) accounted for over 60 percent of the total additional deaths in the first two years of the pandemic.

Data taken from: THE WORLD BANK

The impact of COVID-19 - Covid-19 Vaccination

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Vaccination to combat the Covid-19 pandemic

In this graph we can see the importance of the vaccination process as a way to combat the pandemic, because when the number of vaccines per person exceeds one vaccine, the number of deaths from Covid- 19 begins to decline leading to the end of the pandemic, declared in May 2023 by the WHO.

Data taken from: Our World in Data | THE WORLD BANK

Conclusion

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Increase in average life expectancy

The development of medicine to combat and prevent diseases has led to an increase in average life expectancy around the world.



Fight against communicable diseases

Many countries are already close to ending the epidemics of AIDS, tuberculosis, malaria and neglected tropical diseases and combating hepatitis, waterborne diseases and other communicable diseases, Africa is the continent that is furthest away.



Impact of Covid-19

Covid-19 has affected populations very severely, increasing mortality rates across the world and generally worsening many benchmark health indicators. Only the strong vaccination policy implemented by most countries managed to stop the pandemic and prevent new deaths from the disease.